

Yuming Gu

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EDUCATION

University of Southern California <i>Ph.D in Computer Science advised by Prof. Stefanos Nikolaidis and Prof. Hao Li</i>	Los Angeles, CA Sep. 2021 – Now
University of Southern California <i>M.S. in Computer Science advised by Prof. Hao Li</i>	Los Angeles, CA Aug. 2019 – Dec 2021
University of California, Berkeley <i>Exchange Student</i>	Berkeley, CA July 2018– Aug. 2018
Dalian Maritime University <i>B.S. in Electrical Engineering</i>	Dalian, China Aug. 2015 – May 2019

HONORS & AWARDS

<i>GSG Professional Development Fund, University of Southern California.</i>	2024
<i>First Award Winner, Mathematical Contest in Modeling (MCM/ICM), COMAP.</i>	2019
<i>National Scholarship.</i>	2017, 2018

RESEARCH PUBLICATIONS & PREPRINTS

ENVISION: Embodied Visual Planning via Goal-Imagery Video Diffusion Yuming Gu, Yizhi Wang, Yining Hong, Yipeng Gao, Jiang Hao, Angtian Wang, Bo Liu, Nathaniel S. Dennler, Zhengfei Kuang, Hao Li, Gordon Wetzstein, Chongyang Ma Under Review, RSS 2026	Page
Condition Matters in Full-head 3D GANs Heyuan Li, Huimin Zhang, Yuda Qiu, Zhengwentai Sun, Keru Zheng, Lingteng Qiu, Peihao Li, Qi Zuo, Ce Chen, Yujian Zheng, Yuming Gu, Zilong Dong, Xiaoguang Han International Conference on Learning Representations (ICLR), 2026	Page
DiffPortrait360: Consistent Portrait Diffusion for 360 View Synthesis Yuming Gu, Phong Tran, Yujian Zheng, Hongyi Xu, Heyuan Li, Adilbek Karmanov, Hao Li IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2025	Page
DiffPortrait3D: Controllable Diffusion for Zero-Shot Portrait View Synthesis Yuming Gu, Hongyi Xu, You Xie, Guoxian Song, Yichun Shi, Di Chang, Jing Yang, Linjie Luo IEEE Conference on Computer Vision and Pattern Recognition (CVPR Highlight Top 9.3%), 2024	Page
DisUnknown: Distilling Unknown Factors for Disentanglement Learning Sitao Xiang, Yuming Gu, Pengda Xiang, Menglei Chai, Hao Li, Yajie Zhao, Mingming He IEEE International Conference on Computer Vision (ICCV), 2021.	Page
One-Shot Identity-Preserving Portrait Reenactment Sitao Xiang, Yuming Gu, Pengda Xiang, Mingming He, Koki Nagano, Haiwei Chen, Hao Li arXiv 2020	
Protecting World Leaders Against Deep Fakes Shruti Agarwal, Hany Farid, Yuming Gu, Mingming He, Koki Nagano, Hao Li IEEE Conference on Computer Vision and Pattern Recognition (CVPR) Workshops, 2019.	

EXPERIENCE

TikTok Ltd <i>Research Intern</i> Working on a Research Project on Foundational Video Generation Model with Dr. Chongyang Ma	San Jose, CA May 2025 – Current
MBZUAI <i>Visiting Student</i> Research Project on Full Head Novel View Synthesis	Abu Dhabi, UAE May 2024 – Aug 2024

TikTok Ltd

Research Intern

San Jose, CA

May 2023 – Nov 2023

- Working on Single Image portrait novel view synthesis with **Dr. Hongyi Xu** and **Dr. Linjie Luo**

University of Southern California

Research Assistant

Los Angeles, CA

June 2019 – Present

- Worked on AI Driven face synthesis, Conversational AI project with **Prof. Hao Li**
- Diffusion Research with **Prof. Stefanos Nikolaidis**
- Processed Lightstage data to 3D mesh with Unified Topology

Professional Reviewer Service

CVPR, ICCV, ECCV, Neurips, ACM MultiMedia, WACV, 3DV

TEACHING EXPERIENCE

Teaching Assistant, CSCI585: Database System

Fall 2022, 2023

Teaching Assistant, CSCI576: MultiMedia System

Fall 2025

VOLUNTEER ACTIVITIES

The World Economic Forum Davos, 2017,2019. ACM Programming Contest 2019